

Recycling of Single-Use Multilayer Plastics for Biomanufacturing with Solvent-Targeted Recovery and Precipitation

Tushar, Charles Granger, Ali Altamimi, Yoel Cortes-Pena, Kevin Nelson, Xiaobo Dong, Karl Britt, Louis Barrows, Andrea Thurnheer, Styliani Avraamidou, Reid C. Van Lehn, and George W. Huber*



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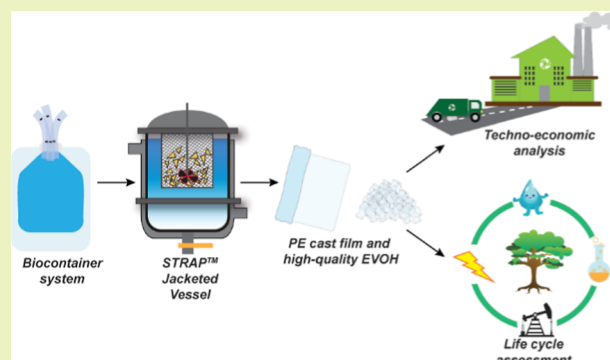
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ABSTRACT: This study demonstrates the recycling of single-use technology (SUT) plastics for biomanufacturing by solvent-targeted recovery and precipitation (STRAP), which recovers polymer resins by selective dissolution and precipitation. Polyethylene (PE) and ethylene vinyl alcohol (EVOH) were extracted from the films of unused biocontainer bags. Our findings demonstrate that a cast film from STRAP-recovered PE exhibits comparable mechanical properties and color to a cast film from virgin LDPE, validating its potential for practical reuse in film manufacturing. Life cycle analysis shows STRAP consumes 50% less fossil fuel and produces 50% fewer greenhouse gas emissions than virgin PE production and achieves even greater reductions—80% decrease in both fossil fuel use and emissions—compared to virgin EVOH production. Economic analysis reveals that, due to the high value of EVOH, a STRAP plant processing a feed of 90% PE and 10% EVOH could be profitable at a scale as small as 3 kton/year. The separation of EVOH greatly increases the profitability of a recycling facility, compared to facilities only able to recover PE, while a plant that only recovers PE needs at least nearly 9 kton/year to be profitable. This work demonstrates that STRAP could be a viable step toward achieving circularity in the biomanufacturing industry by minimizing greenhouse gas emissions and reducing the reliance on incineration and landfilling for SUT disposal.

KEYWORDS: single-use technology, plastic recycling, dissolution, techno-economic analysis, life cycle assessment



INTRODUCTION

Single-use technologies (SUTs) have revolutionized the biopharmaceutical industry compared to traditional stainless-steel technologies. SUTs are presterilized disposable components (bags, filters, tubing, mixers, storage bags) used in biomanufacturing. SUTs provide increased flexibility and productivity, faster time to market, decreased utility consumption, and reduced costs compared to traditional stainless-steel equipment. SUTs reduce the risk of cross-contamination, thereby enhancing sterility.¹ Closed system processing also lowers facility sterilization requirements, thereby reducing the facility's energy consumption. SUTs are typically composed of multilayer polymer films, tubing, bioreactor bags, magnets, filters, sensors, and other disposable components. Despite the sustainability benefits of SUTs, the use of SUTs generates large amounts of waste. Millipore Sigma estimates that the biopharma industry disposes of 94,000–200,000 tons of plastics every year,² including 30,000 tons of plastics from SUTs.^{3,4} Due to stringent regulations in the biopharma industry, plastics in SUTs are high-purity, containing very few contaminants and additives. Used SUTs are commonly treated as hazardous waste and perceived as difficult to recycle.⁵ Consequently, SUTs are typically incinerated or

landfilled after a single use, contributing to greenhouse gas (GHG) emissions,⁶ which motivates recycling efforts.

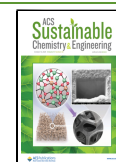
Despite numerous efforts to apply circularity to the use of SUTs, there is limited technology and infrastructure available at a scale to effectively recover and reuse valuable components of SUTs. To date, a limited number of studies have attempted to mechanically recycle SUT virgin materials. In one case, Luu et al.⁵ mechanically recycled unused SUT bags made of polyethylene (PE) and ethylene vinyl alcohol (EVOH). While the recovered PE exhibited mechanical properties comparable to virgin PE, the EVOH was not recovered and the pellets after extrusion were yellow. In another study, Barbaroux et al.⁷ mechanically recycled a small-scale plastic-based bioreactor made of polycarbonate. The recycled polycarbonate had mechanical properties comparable to those of virgin polycarbonate and can be reused in the same application as virgin

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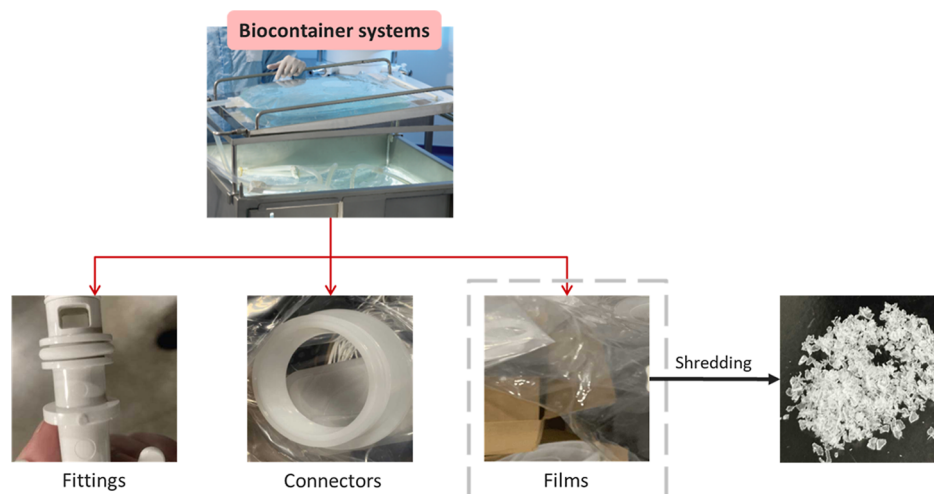


Figure 1. Cytiva biocontainer systems consisting of fittings, connectors, and films. Shredded films were chosen as the feedstock for STRAP in this study.

polycarbonate in cell culture applications. However, the recycled polycarbonate was also yellow. The yellow color is possibly caused by heat, oxygen, or UV light breaking down additives or polymer chains.⁸ Yellowing is similarly observed during polymer aging or weathering and mostly affects the surface⁹ without altering the mechanical properties; for example, Kim et al.¹⁰ found that the key mechanical properties of polycarbonate, polypropylene, and high-density polyethylene stayed stable even after four years of outdoor exposure. The poor quality of the materials obtained from mechanical recycling motivates research into alternative recycling methods for SUTs.

This paper will discuss the recycling of multilayer films from SUTs using solvent-targeted recovery and precipitation (STRAP).^{9–17} STRAP is a solvent-based recycling method that uses solvents to selectively dissolve and precipitate a specific polymer from polymer mixtures without dissolving other polymers.¹³ This method has been demonstrated with multilayer plastic films under relatively mild conditions.^{11,13,18} Solvent-based recycling methods like STRAP can be applied to a variety of different feedstocks, such as printed multilayer films and agriculture films, and have been shown to produce higher-quality recycled polymers compared to mechanical recycling.^{11,20} Solvent-based recycling is advantageous compared to mechanical recycling because it can remove the additives, inorganic and organic, including halogens, sulfur-containing compounds, and antioxidants, as well as inks.^{21–24} Solvent-based recycling is also a viable method to remove individual plastics from complex composite materials and mixed waste streams.^{25–30} In comparison, mechanical recycling is challenging for contaminated and mixed plastics, and it can cause thermo-mechanical degradation.³¹ Recycling plastics using pyrolysis is being used commercially for polyolefin-rich feedstocks at large scales (above 30 kton/year) when integrated with a steam cracker or other petrochemical complex.³² This paper presents the results of STRAP technology applied to the recycling of unused SUT multilayer plastic bags. We show that STRAP yields a clear, high-quality PE film without any visible yellow color. We also assess the economic and environmental impacts of using STRAP to recycle SUT materials.

METHODS

Materials. Unused “single-use biocontainer systems” were collected from Cytiva. As shown in Figure 1, each system had three components: films (made of PE and EVOH), connectors, and fittings. The films contain approximately 90% PE and 8% EVOH. Each component was shredded to 3 mm in size through a 1/8 in. cross-cut shredder (Make: Allegheny 16–75 CX) at Michigan Technological University. For this study, we only used films as feedstocks for STRAP because they were the major component by mass. Connectors and fittings will be analyzed in later studies. Virgin pellets of LDPE (Westlake EG412), mLLDPE (ExxonMobil Exact 3139), and EVOH (Soarnol DC3203RB) were provided by Amcor. Dodecane, a mixture of isomers, was received from Thermo Scientific Chemicals. Dimethyl sulfoxide (DMSO) (ACS reagent, ≥99.5%) was received from Sigma-Aldrich. Heptane, toluene, and xylenes were received from VWR Chemicals.

Computational Modeling. Solvent screening for the selective dissolution and recovery of PE and EVOH was accomplished following the approach developed in our previous work.^{11,15} Polymer solubilities in various solvents were predicted using the conductor-like screening model for realistic solvents (COSMO-RS), a combined quantum chemical and statistical mechanical approach to calculate temperature-dependent polymer solubilities in various solvent systems.^{15,33–35} As shown in Figure S1, the COSMO-RS represents each molecule based on the screening charge density, obtained from density functional theory calculations in our previous work, that arises at molecular surfaces due to polarization of the surrounding medium. We approximated the screening charge densities of polymer molecules by modeling oligomer structures with deactivated terminal groups. COSMO-RS solubility calculations were performed using COSMOtherm 19 with the BP_TZVP 19 parametrization. The workflow established in our previous work can be adapted by using different reference inputs so that resin-specific polymer properties can be addressed with a minimum number of calculations, but resin information was not available for this commercial feedstock. Therefore, LDPE calculations relied on our previous PE molecular model with a reference melting temperature of 113 °C and reference solubility of 23.1 wt % at 110 °C in toluene, and with 31 conformers of a 6-mer structure for molecular modeling. Similarly, the EVOH calculations used previous experimental values with a reference melting temperature of 177.47 °C and a reference solubility of 19.4 wt % in DMSO at 95 °C, with 24 conformers of a 6-mer (4VA:2E) random copolymer structure for molecular modeling. The feedstock also contained a small percentage of a grafting layer, which consists of LDPE covalently bound to EVOH through a ring-opened maleic anhydride constituent. Solubility differences between this grafting layer and pure LDPE and EVOH were negligible due to the low wt %

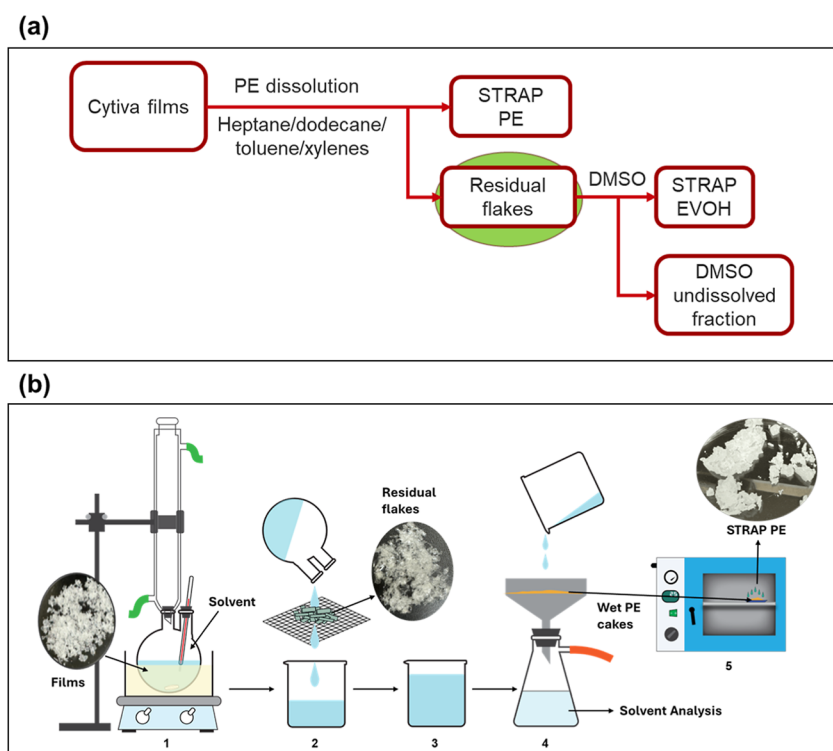


Figure 2. (a) Process flow diagram for PE and EVOH recovery and (b) schematic of the small-scale STRAP setup for PE recovery. Step 1: plastic dissolution, step 2: hot filtration, step 3: cooling and precipitation, step 4: cold filtration, and step 5: vacuum drying.

of this grafting layer (<1 wt %), so it was not suitable to extract this component with a selective solvent wash step.

STRAP Experimental Procedures. Polymer recovery using STRAP broadly consists of polymer dissolution, filtration, precipitation, and recovery. STRAP was carried out using two experimental protocols with different heating setups and feed size: (1) a simple setup for a smaller amount of feed (1–20 g) and (2) a jacketed dissolution vessel for a larger amount of feed (150–200 g).¹⁹ Silicone oil was used as heat-transfer fluid to achieve a desired temperature during dissolution.¹⁹

Small-Scale STRAP Setup. Figure 2a shows the overall experimental process for the recovery of PE and EVOH. The setup shown in Figure 2b involves five steps to recover the PE and residual film components. In step 1, 5 g of shredded films is dissolved in 100 g of dodecane (feed-to-solvent ratio of 1:20) at 110 °C. The solution is stirred in a silicone oil bath with a magnetic stirrer for 90 min to maximize PE dissolution. In step 2, the hot solution is filtered out to separate dissolved PE from other components using a stainless-steel wire mesh filter with an opening size of 0.0277 in. and an open area of 44.2%. In step 3, the dodecane–PE solution is cooled at room temperature for 30 min, allowing the PE to precipitate. In step 4, the cooled solution is mechanically filtered under vacuum to recover the precipitated PE. Finally, in step 5, the recovered PE and undissolved residual flakes are dried in a vacuum oven at 100 °C for 2 h. This process yielded 4 g of PE and 1 g of residual flakes for further analysis.

The residual flakes recovered after the dodecane wash were found to be rich in EVOH. The small-scale STRAP setup was applied to 1 g of the residual flakes, and all steps were repeated with 20 g of DMSO as a solvent. After drying in the vacuum oven, this process yielded 0.3 g of EVOH which was 6% of the films by weight. However, some fractions remained undissolved even after the DMSO wash, and there were some material transfer losses as well. The undissolved fraction (residue) was found to be 0.2 g (4% of the films), with the rest of the film mass lost in material transfer.

Jacketed Dissolution Vessel STRAP Setup. A 4 L jacketed dissolution vessel was used to perform experiments at larger scale and produce resins for mechanical testing. The vessel is heated by a

continuous flow of silicone oil through the jacket. The undissolved fractions were collected in a filter mesh made up of steel with a 1000 μm pore size. A total volume of 3.1 L (~2120 g) of heptane was preheated to 95 °C in the vessel before 155 g of shredded film was added, maintaining a feed-to-solvent ratio of 1:13.6. This feed-to-solvent ratio was achieved after averaging 7–8 runs. A feed-to-solvent ratio of 1:12 resulted in a low PE yield (70%) and undissolved flakes (20%) that were contaminated with PE. Undissolved flakes, presumed to be EVOH, were separated using an internal filter mesh and retained for subsequent analysis, as detailed later in this study. To generate sufficient STRAP PE for a cast film, the experiment was performed in ten iterations, with the average batch material balance illustrated in Figure 3.

Characterization. *FTIR.* Attenuated total reflectance Fourier-transform infrared (ATR-FTIR) spectra were acquired with a Bruker Vertex 70 FTIR spectrometer equipped with a liquid-nitrogen-cooled mercury cadmium–telluride (MCT) detector (Bruker Corporation, Billerica, Massachusetts, USA). An average of 16 scans was collected in the range of 4000–600 cm^{-1} with a resolution of 4 cm^{-1} . Each spectrum was scanned with an air background.

TGA. Thermogravimetric analysis (TGA) was conducted on a TGA Q500 analyzer (TA Instruments, New Castle, Delaware, USA) from 40 to 550 °C for PE with a ramping rate of 15 °C min^{-1} under a nitrogen flow rate of 50.0 mL min^{-1} . For EVOH, TGA was conducted on the same machine from 40 to 220 °C with a ramping rate of 15 °C min^{-1} and held at 220 °C for 10 min.

DSC. Differential scanning calorimetry (DSC) was used to estimate the melting temperature (T_m) and crystallization temperature (T_c) on a DSC Q100 calorimeter equipped with a Refrigerated Cooling System 90 (TA Instruments, New Castle, Delaware, USA). Samples were encapsulated in standard aluminum pans, heated from –20 to 220 °C with a ramping rate of 15 °C min^{-1} and cooled from 220 to –20 °C at the same ramping rate. The heating/cooling process was looped twice at a 50.0 mL min^{-1} nitrogen flow rate.

GC–MS. Gas chromatography–mass spectrometry (GC–MS) was used to analyze the solvents on a Shimadzu GCMS-QP2010 mass

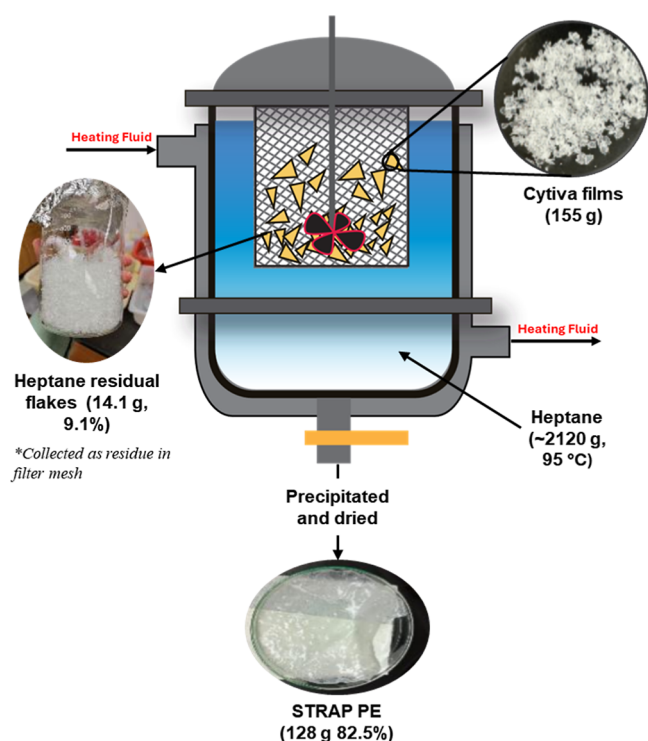


Figure 3. Schematic of the jacketed dissolution vessel for large-scale STRAP experiments.

spectrometer with an RTX-VMS column and quadrupole MS (Shimadzu Corporation, Kyoto, Japan).

Hot Press for Compressed Films. Small films for EVOH testing were produced by placing 0.75 g of polymer between two 6 in. × 6 in. PTFE-coated fiberglass sheets followed by two 6 in. × 6 in. aluminum sheets on the outside. The Carver Press 3970 was used to heat the sample at 190 °C without pressure for 1 min, followed by 3 mt of pressure for 1 min.

Extrusion of Recovered Polyethylene and Cast Film Production. The PE recovered from the jacketed vessel STRAP method was cryoground with liquid nitrogen in a Thomas Model 4 Wiley Mill to 3 mm granules before extrusion. The PE powder was extruded to 3 mm pellets using a Leistritz twin-screw extruder with a temperature profile of 170–190 °C and a 19 rpm screw speed. After

this, the PE pellets were extruded into a cast film at an average barrel temperature of 177 °C and a die temperature of 193 °C with a Labtech Engineering cast film extruder.

Characterization of the PE Cast Film. Tensile properties of the films were measured using an Instron testing machine following ASTM Standard D882-1.³⁶ The thermal properties of the films were analyzed with a NETZSCH TGA 209F1Libra. The analysis compared the dynamic cooling and the second heating of the films, from −20 to 250 °C, at rate of 10 K/min. Final color of the cast films was quantified with a spectrophotometer and measurements were converted to CIELAB color space values, L^* (lightness), a^* (red-green color component) and b^* (blue-yellow color component), and YI (yellowness index).³⁷ Optical properties were analyzed following the ASTM standard D1003-21 for haze and luminous transmittance of transparent plastics.³⁸ The defect analysis for determining the number of impurities in the cast films was performed with an Amcor internal method.

Technoeconomic Analysis (TEA) and Life Cycle Assessment (LCA). The process model was developed in BioSTEAM, an open-source platform in Python for the design, simulation, and evaluation (via TEA and LCA) of chemical processes.^{39,40} The model is based on data from laboratory experiments, the construction of a 25 kg/h STRAP pilot plant at Michigan Technological University,⁴¹ and builds upon past TEA/LCA studies on STRAP.^{13,17}

The plastic feedstock is fed into a dissolution vessel and is mixed with the solvent to dissolve the target polymer (Figure 4). The effluent is sent through a series of filters to separate the solids (i.e., insoluble plastics and biogenic material) from the solution. A vacuum twin-screw drier recovers >99.9% of the solvent retained on the undissolved plastic. The undissolved plastic can be landfilled or further processed to remove another resin layer. The dissolved resin is then precipitated by cooling within a jacketed precipitation and filtration module. After precipitation, a melt devolatilizer removes the solvent down to <1 wt %, and an outgassing extruder recovers the last solvent fraction, leaving low concentrations (<10 ppm) in the resin. Plastic additives, inks, and any soluble biogenic material which may accumulate in the solvent are removed by adsorption with granulated activated carbon. The design of the adsorption module follows the procedure outlined by Gabelman.⁴² The solute capacity was estimated using the fitted Freundlich isotherm of Yellow 12 dye (commonly found in plastics) with activated carbon.⁴³ The length of the unused bed was estimated through numerical mass transfer modeling. After adsorption, the cold solvent is reheated and returned to the dissolution tank. To improve energy efficiency and economics, a heat exchanger was added between the hot and cold solvent streams.

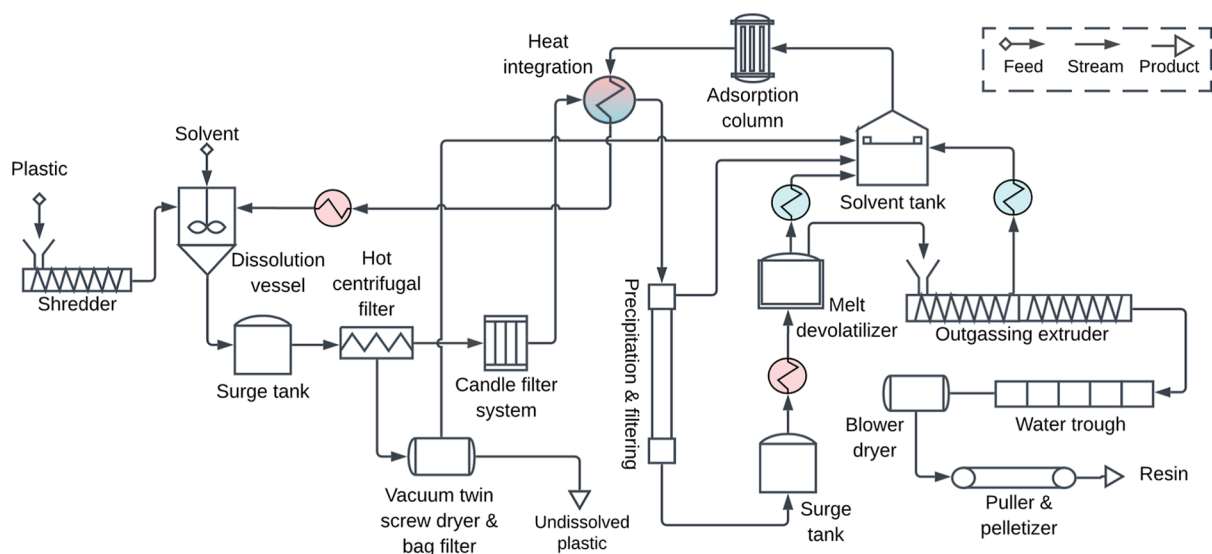


Figure 4. Simplified process flow diagram of a single wash cycle for the STRAP process.

This also makes the process less sensitive to low precipitation and high dissolution temperatures.

Techno-Economic Analysis. Cost estimates for solvent removal equipment were provided by Amcor. Cost estimates for specialized STRAP equipment (e.g., dissolution and precipitation vessels) were provided by industry vendors. The 6/10th rule for economies of scale was used to formulate capital cost correlations for the specialized modules in STRAP.⁴⁴ The cost of the adsorption columns was estimated as pressure vessels using ASME guidelines to vessel thickness.⁴⁴ All other equipment cost correlations originate from various public sources,^{44,45} as detailed by Cortes-Pena and co-workers.³⁹ The chemical engineering plant cost index (used to account for the increasing cost of equipment due to inflation) was assumed to be 816.0. All parameters used in the discounted cash flow analysis follow the assumptions made in NREL's cellulosic ethanol report.⁴⁶

Life Cycle Assessment. The system boundary extends from the cradle to facility gate for the operational phase and does not include the construction phase. None of the environmental impact was allocated to plastic sent to landfill, in agreement with past LCA studies.¹⁴ The environmental impact of multiple coproduct resins was allocated by mass. All operational inventory data (i.e., life cycle inventory) originated from the model leveraged in this study. The life cycle impact assessment methodology used was the Intergovernmental Panel on Climate Change (IPCC) 2013,⁴⁷ with a focus on GWP₁₀₀ as the primary indicator. Life cycle inventory data for virgin resins, raw materials, and ancillary inputs were adapted from the Environmental Footprint 2.0 database.

RESULTS AND DISCUSSION

Development of STRAP Processing Steps. The general steps for the development and assessment of a new STRAP process for a multilayer film include identifying the polymer components of the film, choosing solvents and operating temperatures suitable for selectively dissolving each polymer component, performing STRAP at a small scale to assess recovered resin yields and properties, and then repeating STRAP at a larger scale to evaluate the properties of films created from STRAP resins. We first acquired the FTIR spectra of the shredded films to verify the composition of polymers in the film. The FTIR spectra of the Cytiva film and virgin LDPE (Figure S2) were compared and display identical spectral bands with no bands belonging to other resins. This comparison confirmed that the films mainly consisted of PE, with EVOH expected as a minority component, as will be discussed later in the paper.

To choose STRAP solvents and operating temperatures, we leveraged our computational modeling approach to identify solvents with high predicted solubility for the target polymer (PE or EVOH) at an elevated temperature and negligible solubility for the other polymer. Additionally, ideal solvents have a low solubility for both polymers at room temperature, allowing for temperature-based precipitation. Figure S1c lists example solvents that are selective for each polymer, along with predicted solubilities. Based on these criteria, xylene, dodecane, toluene, and heptane were identified as potential selective solvents for LDPE dissolution. Similarly, DMSO, ethylene glycol, and propylene glycol were identified as potential selective solvents for EVOH dissolution. These solvents are commonly available and were selected from a larger database of 1000+ solvents. For initial small-scale experiments, we selected dodecane as the selective solvent for PE because of its low volatility, thereby decreasing unintentional solvent loss in this laboratory setup, and DMSO as a selective solvent for

EVOH because it is easy to work with due to its low viscosity and has been used to dissolve EVOH in previous work.^{13–15}

Small-Scale Polymer Extraction from Films via STRAP. We applied the small-scale STRAP experiment (Figure 2b) to extract PE and EVOH from 5 g of the SUT film. Experimental conditions and recovered polymer masses are listed in Table 1. First, PE was extracted by dissolving in

Table 1. Experimental Summary for Small-Scale STRAP Experiments

film component	feed: 5 g		
	solvent used to dissolve the component	recovery (yield)	dissolution time (min)
PE (at 110 °C)	dodecane (100 g)	4.0 g (80.0%)	90
EVOH (at 110 °C)	DMSO (20 g)	0.3 g (6.0%)	90
DMSO undissolved fraction		0.2 g (4.0%)	

dodecane. PE was targeted for extraction first due to its higher abundance in the Cytiva film and because it formed the external layers of the film. Next, EVOH was extracted by dissolving the residual flakes (undissolved fraction after PE dissolution) in DMSO. The total amount of material recovered (i.e., yield) for these experiments was 90% (~4.5 g). The missing mass was lost during material transfer by sticking to the equipment surfaces.

The recovered fractions were analyzed with FTIR to confirm polymer identity and assess purity after each step, as shown in Table 1. Figure 5 shows the FTIR scan of the Cytiva film, the extracted PE (referred to as STRAP PE), the extracted EVOH (referred to as STRAP EVOH), and the remaining DMSO undissolved fraction as well as comparisons to virgin resins. The FTIR results (Figure 5a) show that STRAP PE is comparable to virgin LDPE. Spectra of STRAP EVOH and virgin EVOH (Figure 5b) displayed identical spectral bands without traces of other resins, confirming that the purity of STRAP EVOH is comparable to virgin EVOH. The DMSO undissolved fraction (Figure 5c), after PE dissolution and before EVOH dissolution, showed the presence of PE and EVOH, which will be discussed in later sections.

Further characterization of STRAP PE and STRAP EVOH was done by using thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) to thoroughly analyze the purity of recovered polymers. Comparative TGA, shown in Figure 6, revealed that STRAP PE and virgin LDPE exhibit comparable thermal stability and decomposition behavior. The main decomposition ranges were similar, from 400 to 520 °C.

The DSC curves in Figure 7a and 7b reveal two melting point peaks for Cytiva film around 95 and 113 °C, corresponding to melting points of mLLDPE (98 °C) and virgin LDPE (113 °C), respectively. This indicates that Cytiva films are made up of two different types of polyethylene. The DSC results of the STRAP PE align with those of mLLDPE/LDPE blends (Liu et al.⁴⁸), indicating that STRAP PE is a blend of mLLDPE and LDPE originating from the Cytiva film. Figure S3 and the heat of fusion values in Table S1 estimate the composition of STRAP PE as approximately 20/80 wt % LDPE/mLLDPE. Virgin LDPE exhibits the highest crystallinity, evidenced by its elevated fusion (ΔH_f), while the lower ΔH_f values for STRAP PE and Cytiva film are attributed to

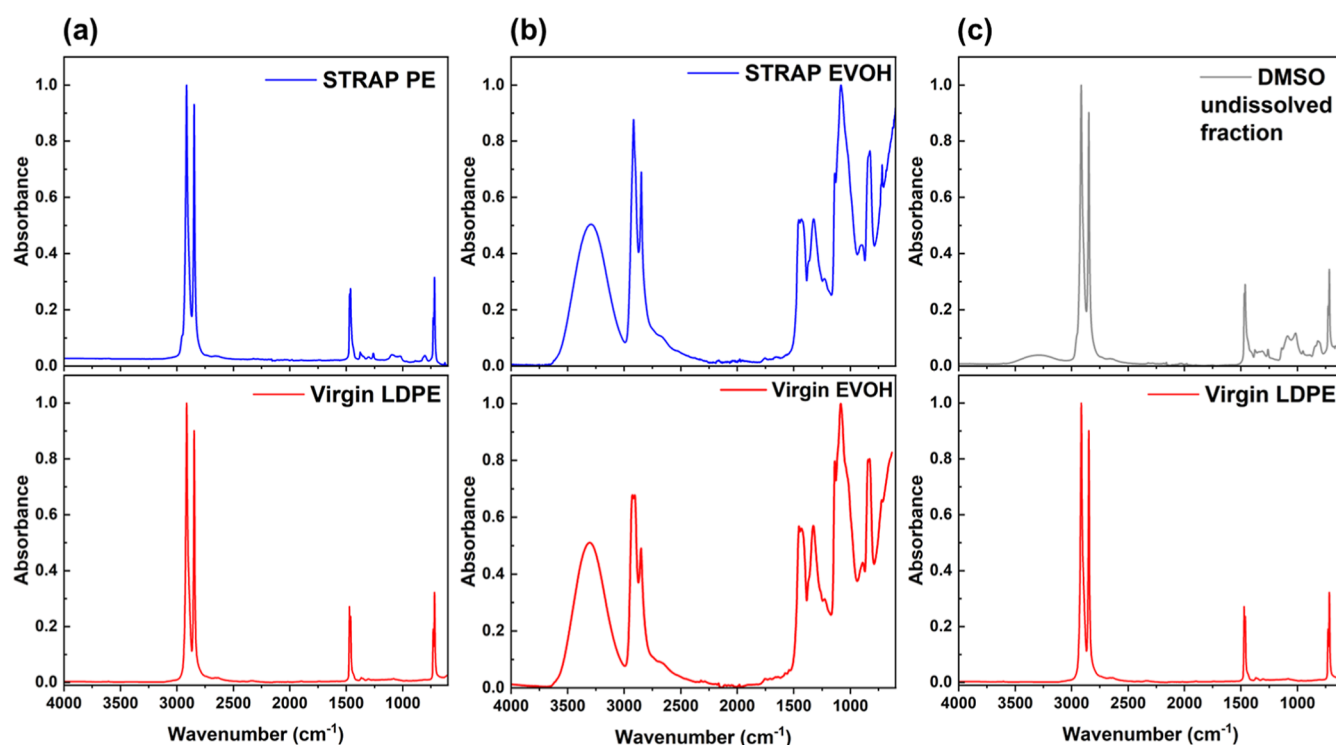


Figure 5. FTIR results of STRAP extracted film components (blue) and virgin resins (red) for (a) STRAP PE (recovered after dodecane wash), (b) STRAP EVOH (recovered after DMSO wash), and (c) DMSO undissolved fraction (after DMSO wash on residual flakes).

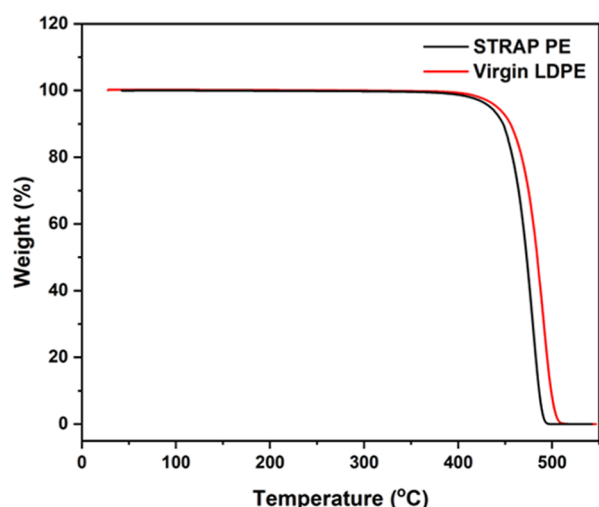


Figure 6. TGA (thermogravimetric analysis) profile of STRAP PE compared to virgin LDPE (ramping rate: 15 °C/min, N₂ flow rate: 50.0 mL/min).

mLLDPE. mLLDPE is classified as plastomers with inherently lower heat of fusion.⁴⁹ The STRAP PE melting point range lies between the melting point ranges of Cytiva film component peaks (95 and 113 °C). This suggests that the extracted material is morphologically different from that of the film. The peak at 193 °C represents the melting peak of EVOH in the film, with a detailed discussion provided in later sections.

Large-Scale Polymer Extraction and Cast Film Production. We prepared cast films of STRAP PE and virgin LDPE for comparison to validate the practical uses of STRAP PE in film manufacturing. Approximately 0.8–1 kg of STRAP PE was required, which was extracted using a jacketed dissolution vessel shown in Figure 3 that can process 150–

400 g of feed per batch, depending upon the bulk density of the feed. For these large-scale STRAP experiments, heptane was chosen as the selective solvent for dissolving PE because its low flash point allowed for easier drying of the recovered polymer.

STRAP PE cast film production involves extrusion and pelletization after drying in a vacuum oven, as shown in Figure 8. The STRAP process was also applied to virgin LDPE pellets (STRAP virgin LDPE) which involved dissolving virgin LDPE pellets in the solvent and recovering the PE after precipitation and drying. This was done to understand the effect of STRAP on the polymer. The pellets were extruded into a cast film, and a third cast film of virgin LDPE (Westlake EG412) resins was also extruded on the same equipment for comparison. Figure 8 shows that the large-scale experiments produce STRAP PE pellets that lead to visually clear, optically transparent, and nonvisible yellow color.

The DSC results of the STRAP PE cast film and virgin LDPE cast film are shown in Figure S4 and Table S1. Table S1 shows that the STRAP PE cast film has low heat of fusion (ΔH_f) which is similar to STRAP PE. This shows the cast film is likely made up of mainly mLLDPE (plastomers) and making cast film did not change thermal properties. Table 2 shows tensile properties of the three films made from STRAP PE, STRAP virgin LDPE, and virgin LDPE. The LDPE control and STRAP virgin LDPE cast films have similar mechanical properties. The STRAP PE film was more flexible than the other two cast films due to its lower Young's modulus, which is likely caused by the presence of mLLDPE in the original feedstock (Figure 7).^{50,51} The low heat of fusion, i.e., low crystallinity, is a characteristic of a plastomer and is responsible for the higher elongation at break and lower Young's Modulus.⁴⁹ It is also possible that some PE oligomers (C10–C15) were permanently removed during STRAP.

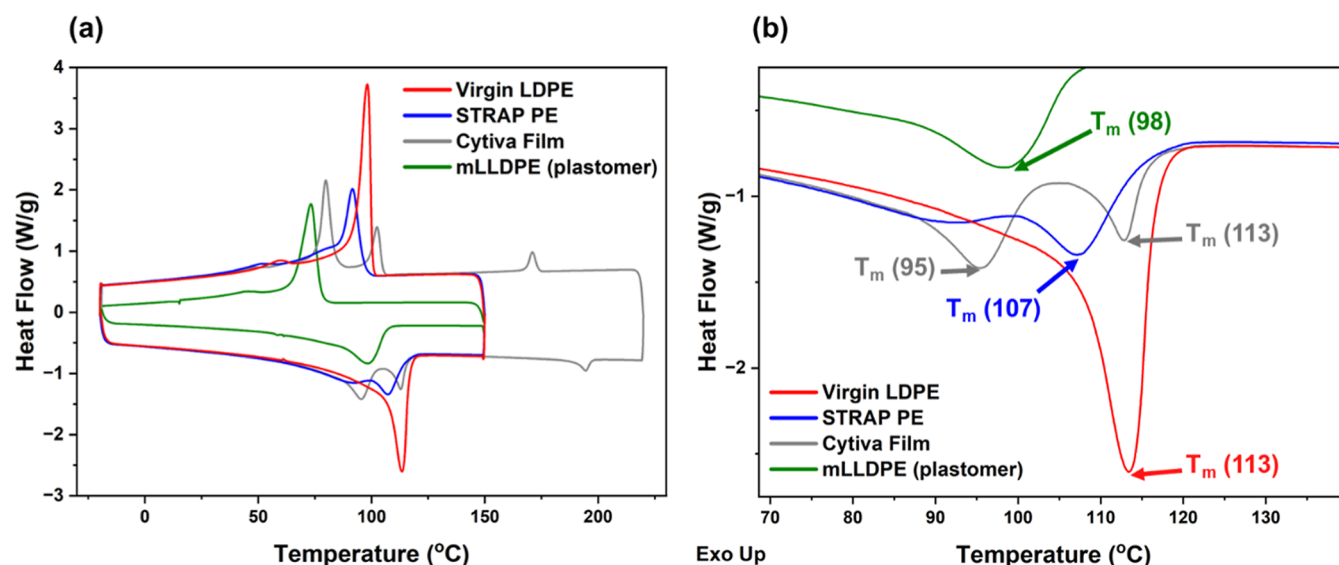


Figure 7. DSC curves (2nd heating cycle) for Cytiva film, STRAP PE, Virgin LDPE, and mLLDPE: (a) complete curves and (b) melting peaks (ramping rate: 15 °C/min, N₂ flow rate: 50.0 mL/min).

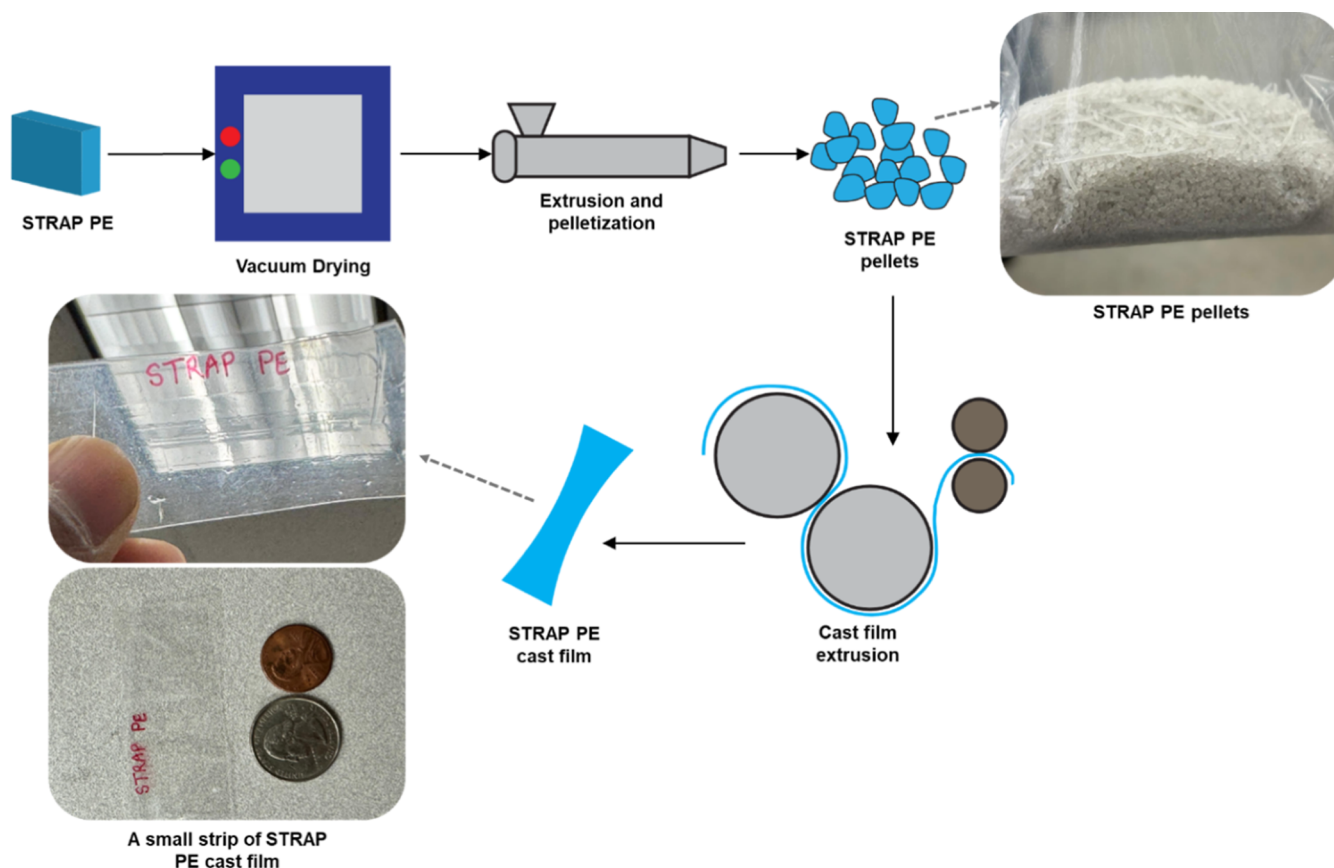


Figure 8. Schematic of cast film production steps. Images show STRAP PE pellets and a small strip of the resulting STRAP PE cast film.

Indeed, the GC–MS analysis from large-scale experiments of the post-STRAP solvents (dodecane, heptane, toluene, and xylenes) shown in Figure S6 confirms the presence of PE oligomers.

In addition to mechanical properties, the visual properties of the films were compared. The CIELAB color test results shown in Figure 9 reveal that the STRAP PE cast film is very similar to the virgin LDPE cast film in color, haze, and transmittance.

From visual inspection, the STRAP PE and STRAP virgin LDPE cast films contained inclusions (“gels”) that were more transparent than those in the virgin LDPE cast film (Figure 9c). These gels are cosmetically unpleasant and can affect optical properties (haze, light scattering) and mechanical properties (film tearing, elongation, impact resistance).⁵² Both STRAP PE and STRAP virgin LDPE were initially extruded and pelletized at the University of Wisconsin, while the LDPE

Table 2. Tensile Strength Properties and Standard Deviation of the STRAP PE Cast Film Compared to the Virgin LDPE Cast Film and STRAP Virgin LDPE Cast Film^{36,a}

sample	thickness (μm)		Young's modulus (MPa)		stress at break (MPa)		force at break (N)		elongation at break (%)	
	MD	TD	MD	TD	MD	TD	MD	TD	MD	TD
STRAP PE cast film	77 $\pm$ 5	78 $\pm$ 1	79.5 $\pm$ 0.9	81.8 $\pm$ 1.1	31.1 $\pm$ 2.0	26.4 $\pm$ 1.5	60.8 $\pm$ 5.9	52.5 $\pm$ 3.3	780 $\pm$ 47	828 $\pm$ 33
LDPE control film	60 $\pm$ 3	61 $\pm$ 1	135.8 $\pm$ 4.7	141 $\pm$ 0.6	25.0 $\pm$ 0.5	18.4 $\pm$ 0.7	38.4 $\pm$ 2.3	28.5 $\pm$ 0.5	493 $\pm$ 34	708 $\pm$ 11
STRAP virgin LDPE cast film	60 $\pm$ 4	60 $\pm$ 1	145.1 $\pm$ 6.4	148.6 $\pm$ 3.7	20.9 $\pm$ 0.9	15.4 $\pm$ 2.3	31.9 $\pm$ 0.7	23.6 $\pm$ 3.6	239 $\pm$ 9	587 $\pm$ 80

^aMD = machine direction, TD = transverse direction.

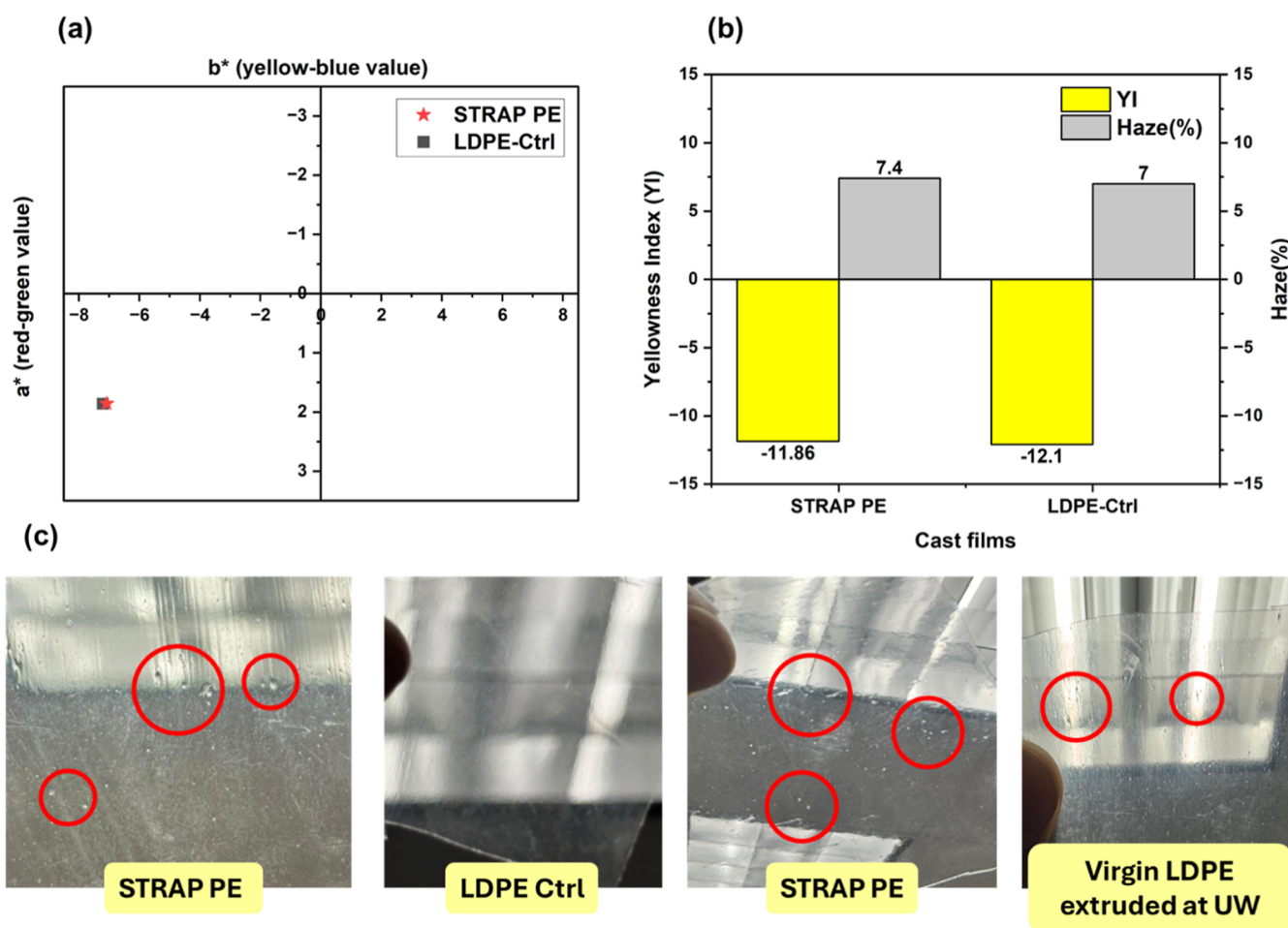


Figure 9. Optical properties of cast films. (a) CIELAB color space values of STRAP PE and LDPE control (LDPE Ctrl) cast films: a^* is the red-green color value ($a^* > 0$ is redder, $a^* < 0$ is greener) and b^* is the yellow-blue color value ($b^* > 0$ yellower, $b^* < 0$ is bluer).⁵³ (b) Yellowness Index (YI) and Haze (%).³⁸ (c) Photographs of different cast films: STRAP PE, LDPE Ctrl, STRAP virgin LDPE, and Virgin LDPE extruded at UW. Gels are circled in red. The LDPE Ctrl film showed no gels.

control was not. The LDPE control film was produced from virgin resins at Amcor using a laboratory-scale extruder, whereas all other films were extruded and pelletized at UW using a shared laboratory extruder. The presence of gels in both STRAP films indicates that the inclusions arose from a processing step rather than from impurities from the initial plastic feed. A fourth cast film was prepared from virgin pellets that were extruded first at the University of Wisconsin and second at Amcor. This fourth film also contained gels (Figure 9c) indicating that the university extruder, not STRAP, was likely the source of contamination. This comparison highlights the impact of extrusion and processing on the film quality.

Optimized Route for EVOH Recovery. Following the positive results with STRAP PE, we aimed to study EVOH recovery. The large-scale jacketed dissolution vessel (Figure 3) method was used to obtain residual flakes after PE dissolution. The aim was to recover high-quality (high EVOH content and low PE content) residual flakes with a single PE-selective solvent. However, the residual flakes recovered from heptane (Figure 10a) during cast film production with a feed-to-solvent ratio of 1:13.6 were found to have visible PE contamination based on comparison with the FTIR spectrum of virgin EVOH (Figure 10b) and the spectrum of the residual flakes recovered from the heptane wash (Figure 10a). The FTIR spectrum in Figure 10a shows that residual flakes have a lower intensity of

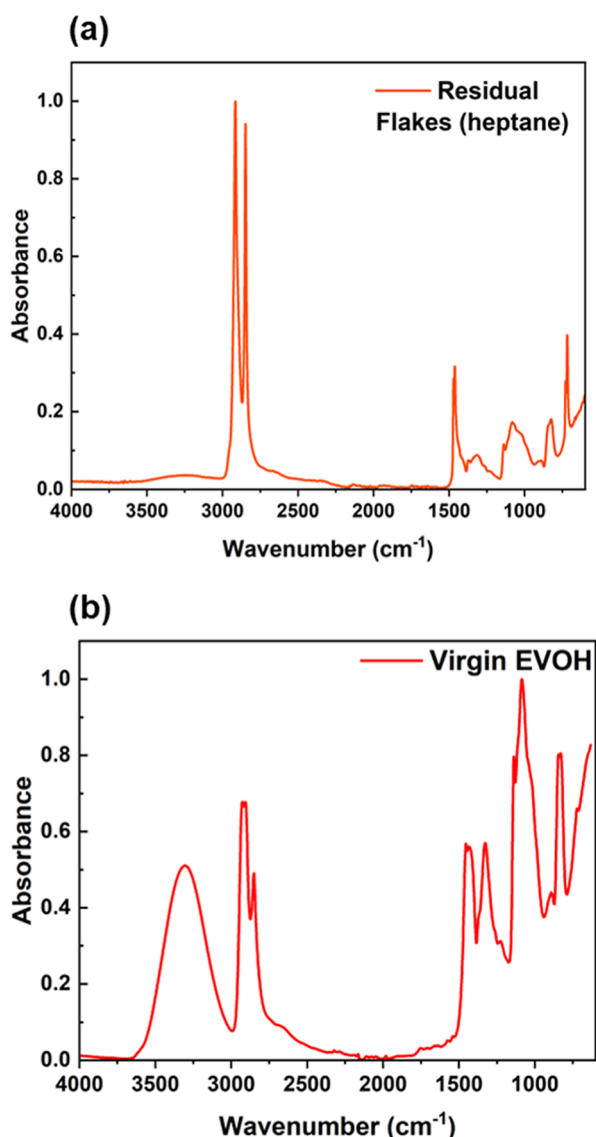


Figure 10. FTIR scans of: (a) Residual flakes (after PE dissolution and before DMSO wash) and (b) virgin EVOH.

OH stretching around 3300 cm^{-1} and clear CH_2 peaks from PE at 720 cm^{-1} , suggesting PE contamination. This led to further investigation of optimal conditions to recover EVOH by dissolving only the PE.

We tested four solvents, heptane, dodecane, toluene, and xylenes, to get high-quality residual flakes comparable to virgin EVOH by dissolving only the PE. Dissolution with heptane was performed at $95\text{ }^\circ\text{C}$, whereas dissolution with dodecane, toluene, and xylenes was performed at $110\text{ }^\circ\text{C}$, all with a consistent feed-to-solvent ratio equal to 1:20. Heptane was heated to $95\text{ }^\circ\text{C}$, slightly below its boiling point ($\sim 100\text{ }^\circ\text{C}$), to minimize vapor pressure buildup and to remain within safe operating limits, considering its low flash point ($-7\text{ }^\circ\text{C}$). Higher temperatures could impair solvent recovery and require design modifications to handle increased pressure in the STRAP systems. Table 3 summarizes these experiments and residual flake recoveries, while Figure 11 compares the FTIR spectra of the residual flakes to virgin EVOH for each of these solvents. Based on FTIR results, there was no noticeable difference in the quality of the EVOH recovered by each solvent. All solvents tested yielded EVOH with minimal

Table 3. Experimental Summary of Large-Scale STRAP Experiments for Residual Flake Recovery

solvent used	feed (in g)	solvent (in g)	PE recovery (%)	residual flake recovery (%)	yield (%)
heptane ($95\text{ }^\circ\text{C}$)	100	2000	85.7	8.8	94.5
dodecane ($110\text{ }^\circ\text{C}$)	100	2000	91.2	7.5	98.7
toluene ($110\text{ }^\circ\text{C}$)	100	2000	88.3	7.8	96.1
xylenes ($110\text{ }^\circ\text{C}$)	100	2000	88.4	7.8	96.2

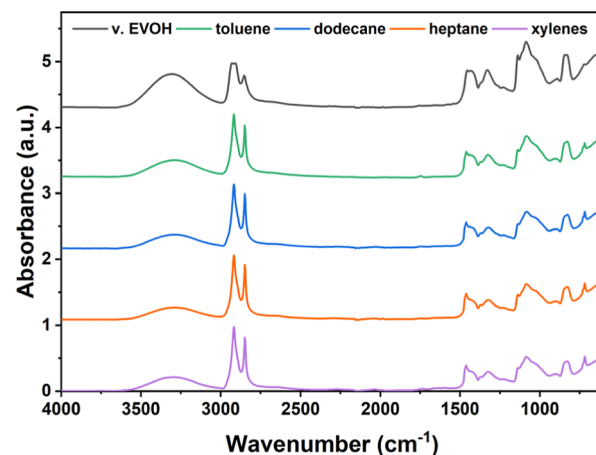


Figure 11. Average FTIR results of residual flakes with feed-to-solvent ratio = 1:20 compared to virgin EVOH.

contamination visible on FTIR. Additionally, Figure S5 shows the FTIR spectra of residual flakes recovered from varied feed-to-solvent ratios. In the cases where less solvent was used, the FTIR spectra also indicated higher PE contamination. These findings provide evidence that the feed-to-solvent ratio can greatly impact the quality of the remaining undissolved polymer, and solvent amount may be more impactful than the type of solvent used.

The variability in the residual flakes after single-solvent washing was assessed by using FTIR, as shown in Figure 12. Each spectrum represents the analysis of 10–15 residual flake samples. Table S2 shows characteristic peaks for PE and EVOH for a better understanding of FTIR spectra of residual flakes. However, variability in the OH stretching intensity ($\sim 3300\text{ cm}^{-1}$) suggests the potential presence of PE adhering to residual flakes, which the solvents failed to remove during the previous step. This may be due to the presence of a coextruded adhesive layer positioned adjacent to the EVOH layer, typically PE grafted with maleic anhydride. The maleic anhydride reacts with EVOH hydroxyl groups to form interfacial covalent bonds.⁵⁴ The maleic anhydride content of a polyolefin tie layer can be 0.15–0.30 wt.% for EVOH adhesion.⁵⁵ However, the maleic anhydride or any other grafting material was not detected with the characterization method used, but it is possible that the quality of residual flakes is affected by it and it is rigorous to detect the anhydride content.⁵⁴

For further characterization, compressed films of xylene-residual flakes were prepared using a hot press at $190\text{ }^\circ\text{C}$. DSC curves of EVOH compressed film and virgin EVOH are shown in Figure 13. The DSC results indicate the relative purity of

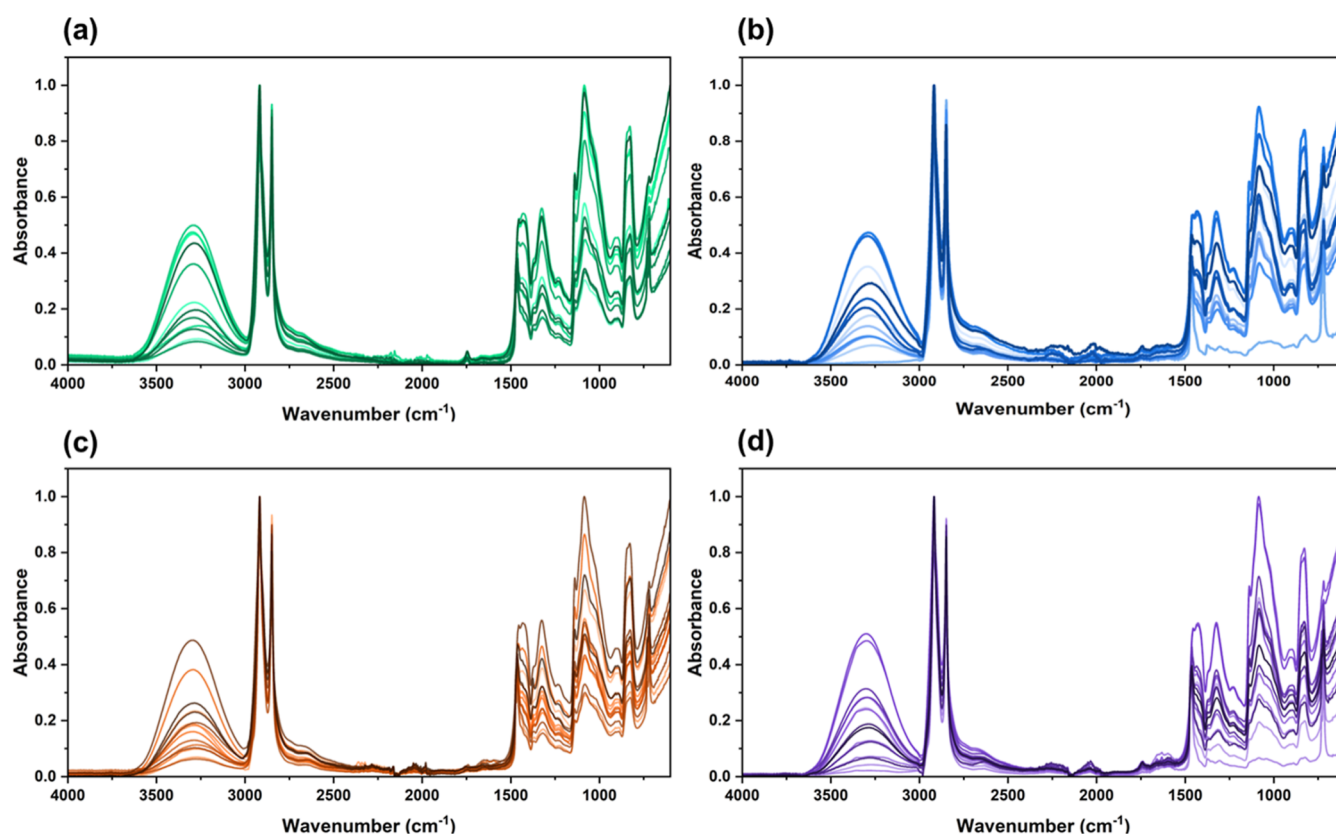


Figure 12. FTIR scan comparison of residual flakes with the feed-to-solvent ratio (1:20), recovered after PE dissolution with (a) toluene, (b) dodecane, (c) heptane, and (d) xylenes.

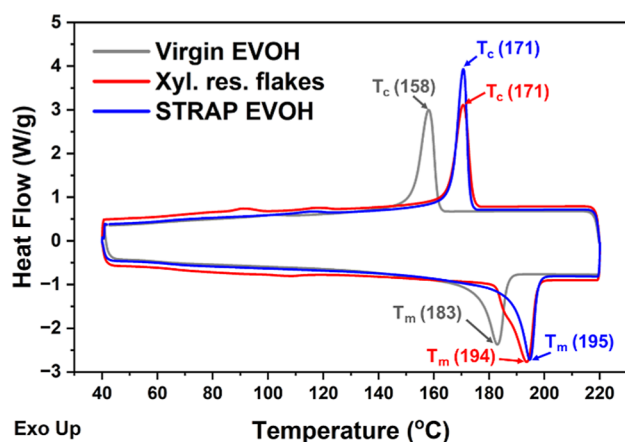


Figure 13. DSC curves (2nd heating cycle) for STRAP EVOH recovered by DMSO dissolution in small-scale STRAP, xylene residual flake compressed film, and virgin EVOH (32 mol % ethylene content); ramping rate: 15 °C/min, N₂ flow rate: 50.0 mL/min.

residual flakes, as their thermal properties align with those of STRAP EVOH, which was the high-quality EVOH recovered after a DMSO wash at a small scale (Figure 5). However, additional testing is necessary to confirm their suitability for commercial use as recycled EVOH.

Techno-Economic Analysis (TEA). The following analysis compares three commercial-scale scenarios for a STRAP plant that processes SUT film waste composed of PE and EVOH as depicted in Figure 14. Additional results are included in workbooks Supporting Information W1, W2, and W3. The SUT waste is received as whole films or bags, and shredding

and preprocessing are included in the model. In scenario 1, only PE is recovered and all undissolved components are landfilled after drying. Scenario 2 is similar but assumes that the undissolved fraction is purified EVOH that can be sold. It uses a single PE-selective solvent, with increased solvent volume compared to scenario 1 to improve residual flake quality, and is best suited for minimally contaminated postindustrial waste. In scenario 3, the undissolved fraction is further processed with an EVOH-selective solvent to purify EVOH. Any remaining solid waste in scenarios 1 or 3 is landfilled at the average California tipping fee of \$72/ton.⁵⁶ Base case parameters for each scenario are listed in Table S3.

The three scenarios are compared in Figure 15, which was created by calculating the product unit operating cost (UOC) per kilogram of given plant capacity. The product is defined as the annual operating costs divided by the annual recycled resin production in kilograms. Also shown is the total capital investment (TCI) for each scenario, which includes all costs required before operation and is further defined in the CAPEX tab of Supporting Information W1, W2, and W3.

Table 4 further compares the 3 scenarios. All values were calculated for a 5 kton/year plant using the base case assumptions from Table S3. The sale prices of the resins were set to match the average prices of virgin PE and EVOH resins, which are \$1.20/kg and \$8.00/kg, respectively. A scale of 5 kton/year was selected for analysis as a representative estimate of collectable SUT film waste in California. Not all scenarios provide a return on investment at 5 kton, so Table 4 also shows the minimum scale required for each process to achieve NPV = 0 with a 15% IRR.

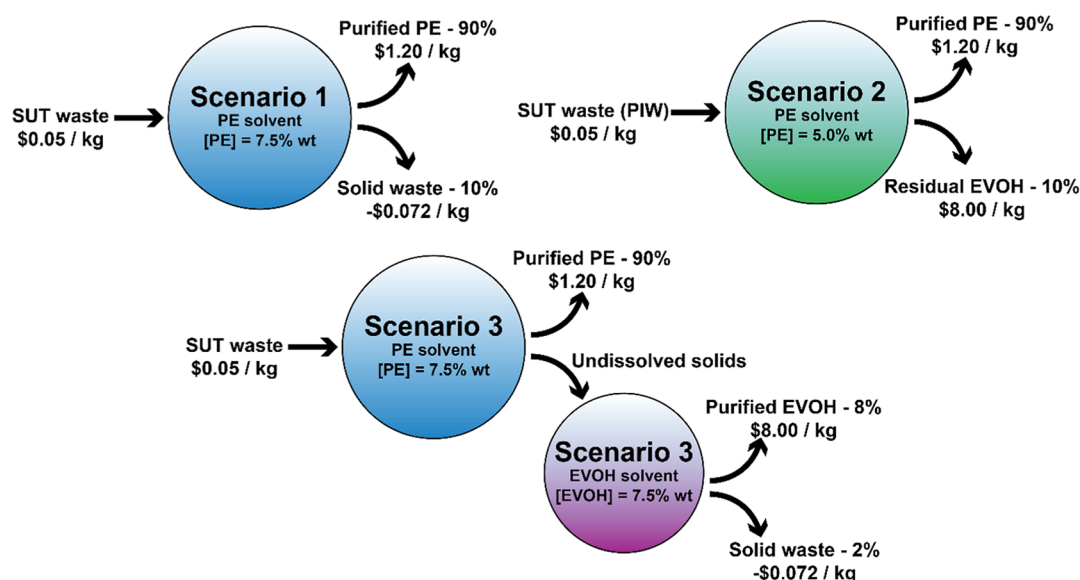


Figure 14. Three base case scenarios for STRAP treatment of PE-EVOH SUT waste. The concentration of the dissolved polymer in solvent is indicated in brackets.

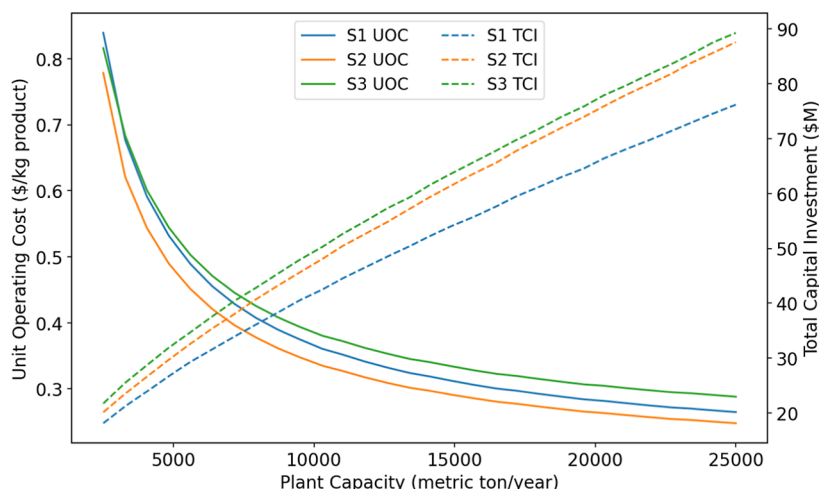


Figure 15. Economic analysis of scenarios 1–3 under base case assumptions including unit operating cost to produce STRAP resin and TCI (ISBL + OSBL + startup costs) vs feedstock capacity.

Table 4. Key Economic Indicators for Each Scenario at 5 kton/year Scale (Base Case) Assuming Resin Price = Virgin Resin Price, IRR = 15%, and PBP = Payback Period^a

economic indicators	scenario 1	scenario 2	scenario 3
TCI	\$27.3 M	\$30.5 M	\$32.3 M
annual revenue (PE/EVOH)	\$5.40 M	\$5.40 M	\$5.40 M
annual operating cost	\$2.33 M	\$2.41 M	\$2.52 M
PBP (years)	13.0	6.38	7.80
min. scale (kton feed/year)	8.66	3.03	4.14
min. scale TCI	\$38.2 M	\$22.4 M	\$29.1 M

^aAlso shown is the minimum scale to allow each scenario to achieve a 15% IRR and the associated TCI.

Scenario 1 has the lowest capital cost but without the added revenue from EVOH, it is the least profitable option. Scenario 2 requires more capital due to increased solvent volume and

equipment size but has the lowest UOC. The recovered EVOH contributes over 40% of scenario 2's revenue, making it the most profitable at 5 kton/year. This scenario depends on a clean, two-polymer feed—likely a well-sorted postindustrial waste—to ensure high-quality EVOH residue. Scenario 3 is the most capital-intensive option and has the highest UOC. However, the purified EVOH product makes this scenario more profitable than scenario 1. Scenario 3 has the potential to produce the highest-quality EVOH, but the economics may suffer if the feedstock regularly contains less than 8% EVOH.

Single-point sensitivity analysis was performed on selected variables to identify the key drivers of economics and environmental impact. All variables, along with their bounds and justifications, can be found in Table S3. Figure 16 displays the most impactful variables on minimum selling price (MSP), for scenarios 1–3. MSP is defined as the minimum product sale price for a plant to be economically viable, covering all costs and the required return on investment over the plant lifetime. MSP was chosen for the sensitivity analysis because it reflects changes in both capital and operating expenses.

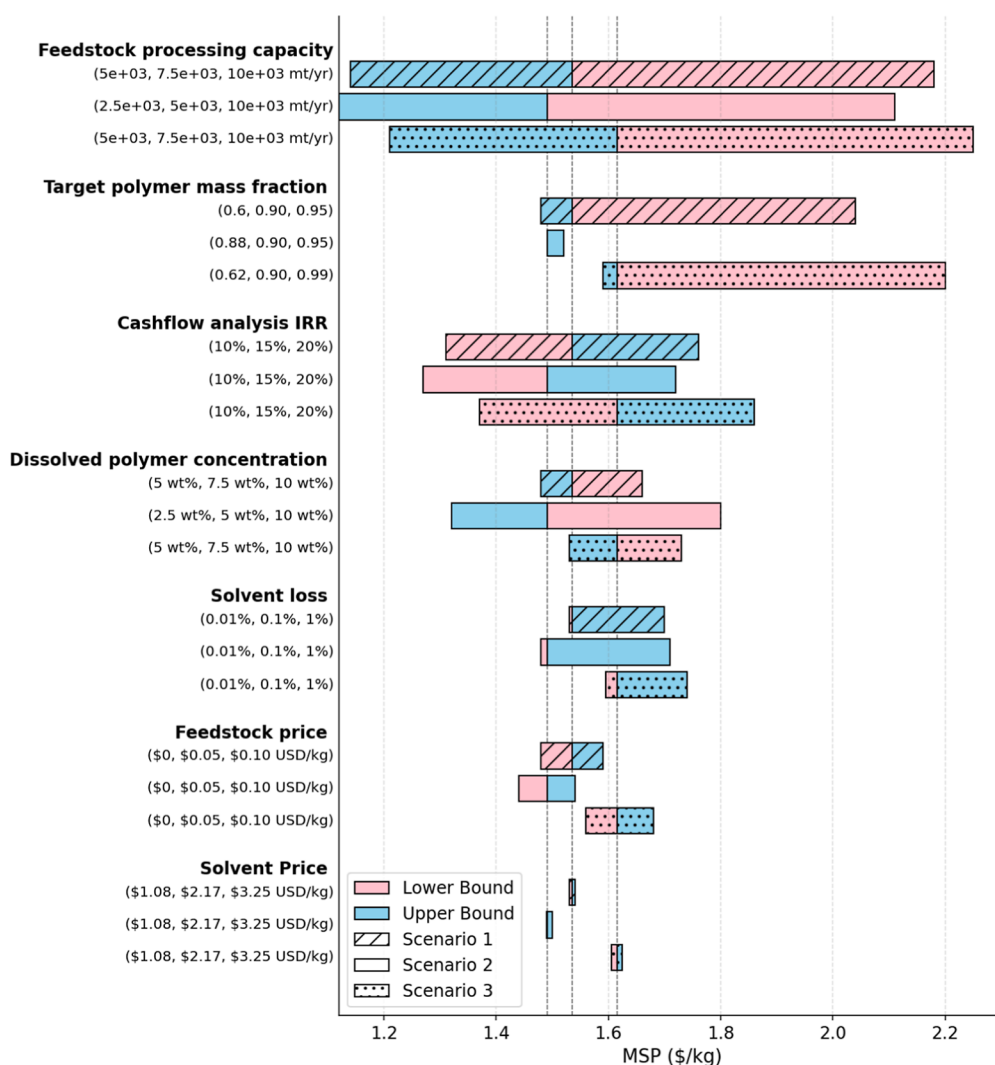


Figure 16. Single-point sensitivity of the most impactful variables on STRAP resin MSP. Variables are defined as the variable name (lower bound, base case, upper bound).

The most important economic driver is processing scale because plant capacity directly affects capital expenses which see benefits from economies of scale. Similarly, dissolved polymer concentration (dissolved solvent capacity) affects required solvent volume and equipment size, directly influencing capital expenses. In scenarios 1 and 3, the target polymer mass fraction in the feedstock strongly influences the economic performance, highlighting the need for careful feedstock selection. In contrast, scenario 2 benefits from a lower target polymer fraction—when two products are recovered with one solvent, less dissolved polymer reduces solvent and equipment requirements. This advantage, however, depends on sourcing clean feedstock that can be separated into two useable products; otherwise, contaminants will be mixed with the undissolved stream. Clean SUT waste and other two-component postindustrial streams are ideal candidates for this scenario.

Additionally, if solvent loss can be limited to 1% or less, it is only the fourth or fifth most impactful variable considered. The plant design includes solvent recovery from the waste and product streams as well as solvent cleaning/regeneration with adsorbents before each recycling step; therefore, maintaining solvent losses below 1% is reasonable.

Ultimately, the commercial design of a STRAP plant that processes SUT waste should be based on the quantity and quality of feedstock available. If sufficiently clean, two-component feed is available, then scenario 2 is a good design. However, further studies are needed to verify if there is a difference in EVOH quality between scenarios 2 and 3 and how that difference may impact the economics. In contrast, scenarios 1 and 3 are more robust options for varied feedstock and are also economically viable above 9 and 4 kton/year, respectively.

Life Cycle Assessment (LCA). Scenarios 1–3 were also subjected to a cradle-to-gate life cycle assessment to directly compare the environmental impacts of the STRAP process to virgin resin production. [Workbook Supporting Information W4](#) includes key assumptions and parameters used to perform the following calculations. [Figure 17](#) shows the global warming potential (GWP) to produce resins in each of the three STRAP scenarios for a plant processing 5 kton/year of SUT waste. STRAP reduces GWP by 58% compared to virgin PE production, and STRAP EVOH emissions are 85% lower in scenario 2 and 73% lower in scenario 3 compared to virgin. Other impact metrics, fossil fuel consumption, water use, human toxicity, ecotoxicity, acidification, ozone depletion, and

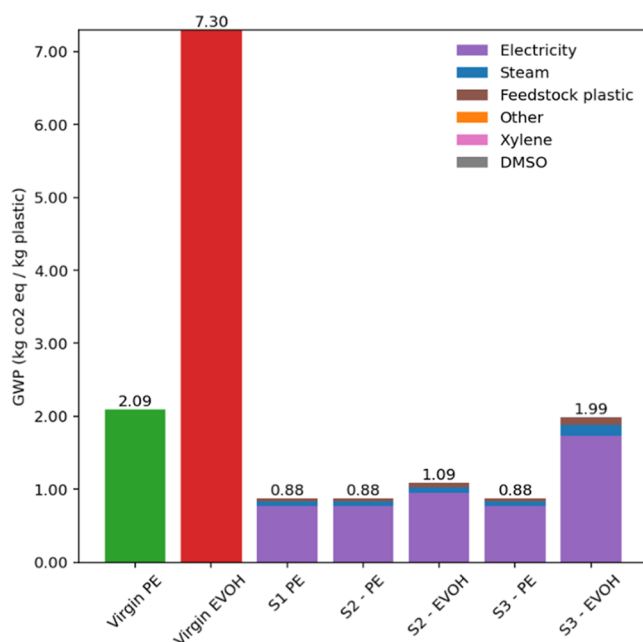


Figure 17. Global warming potential (GWP) for each STRAP scenario.

photochemical ozone creation, are included in Figures S11–S17.

Electricity use is, by far, the main contributor to STRAP's environmental impact, accounting for at least 77% in every impact category analyzed. This study used a representative US grid energy mix, but the STRAP process impacts could be greatly reduced if it was powered by renewable energy sources.

The GWP impacts were also analyzed with a single-point sensitivity analysis (Figure S8). The top four most sensitive variables are in descending order: target polymer mass fraction, feedstock distance, solvent loss, and dissolved polymer concentration. While GWP is not a comprehensive indicator, it effectively highlights relative environmental impact and trends. In all cases, scenario 3 EVOH produces higher impacts than scenario 2, due to the high processing requirements for relatively little product. This, again, emphasizes the importance of scale and feedstock selection.

CONCLUSIONS

STRAP successfully extracted high-quality polyethylene (PE) and ethylene-vinyl alcohol (EVOH) from single-use technology (SUT) biocontainer films. The resulting STRAP PE cast film exhibited comparable color and mechanical properties to those of a virgin low-density polyethylene (LDPE) cast film. However, the STRAP PE cast film displayed a lower Young's modulus, mainly attributed to the presence of mLLDPE plastomer in the Cytiva films. The STRAP PE was mainly composed of mLLDPE. STRAP also yielded high-quality residual EVOH flakes in a single PE-solvent wash. The quality of the residual EVOH flakes increased with increasing solvent amounts.

The techno-economic analysis shows that capital expenses and feedstock composition and quality are major drivers of the profitability of a commercial STRAP plant that processes SUT waste. Recovering EVOH can increase the revenue of a plant by nearly 40%. However, the capital cost of a plant that recovers EVOH is higher, which introduces risk if a consistent

feedstock is not available. Additionally, life cycle analysis shows that most configurations of a STRAP plant processing SUT waste have similar environmental impacts while reducing GHG emissions and fossil fuel consumption by 50%. Electricity and heat are the main contributors to STRAP environmental impacts, and these are most influenced by variables such as feedstock composition and solvent loss. Future studies such as oxygen barrier tests on STRAP EVOH and extractable and leachable tests on STRAP resins could provide clearer insights into the value of STRAP-recovered resins. More bioprocess SUTs should be investigated as well to develop a clearer understanding of potential feedstock types and scales.

Our results show that STRAP is a viable recycling method for SUTs that balances economic and ecological sustainability while producing high-quality recovered materials with mechanical, optical, and thermal properties similar to those of virgin resins. However, complex waste streams can contain more than two polymers, requiring further analysis to determine feasible scales for systems with three or more solvents.⁵⁷ As waste complexity increases, a combined approach of mechanical and solvent-based recycling can be a more effective approach to increase recycling rates.⁵⁸ Additionally, pretreatment and sorting require further investigation, as their importance varies with the SUT compositional complexity. Future work will involve comparing thermograms of STRAP solvent blends and melt blends of LDPE and mLLDPE to better interpret the DSC profile of STRAP PE. Further investigation will also determine if the initial application of STRAP to SUTs can facilitate the recovery of additional polymers and valuable materials, like neodymium magnets, PP fittings, filters, and silicone tubing, from other SUT components. While this study focused on postindustrial biopharma plastic waste, future research should assess postconsumer biopharma waste, evaluating its recyclability, quality, and pretreatment requirements compared to post-industrial waste.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.5c06479>.

Computational modeling for PE and EVOH solubility; FTIR results for Cytiva films and residual flakes; DSC curves for different PE discussed in the paper; heat of fusion vs LDPE content; GC–MS results for solvents; thermal properties and FTIR characteristic peaks; base case definitions for each scenario; Monte Carlo simulations for each scenario; single-point sensitivity analysis; and acidification, freshwater ecotoxicity, fossil fuel consumption, human toxicity, and ozone depletion for each scenario (PDF)

Detailed report of PE only (XLSX)

Detailed report of PE EVOH after 1 wash (XLSX)

Detailed report of PE EVOH after 2 washes (XLSX)

LCA data (XLSX)

AUTHOR INFORMATION

Corresponding Author

George W. Huber – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0002-7838-6893; Email: gwhuber@wisc.edu

Authors

Tushar – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States; orcid.org/0009-0008-9047-5515

Charles Granger – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0002-0160-7520

Ali Altamimi – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States

Yoel Cortes-Pena – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States

Kevin Nelson – Amcor, Neenah, Wisconsin 54956, United States

Xiaobo Dong – Cytiva, Marlborough, Massachusetts 01752, United States

Karl Britt – Cytiva, Marlborough, Massachusetts 01752, United States

Louis Barrows – Cytiva, Marlborough, Massachusetts 01752, United States

Andrea Thurnheer – Genentech, South San Francisco, California 94080, United States

Styliani Avraamidou – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States

Reid C. Van Lehn – Department of Chemical and Biological Engineering, University of Wisconsin—Madison, Madison, Wisconsin 53706, United States; orcid.org/0000-0003-4885-6599

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acssuschemeng.5c06479>

Author Contributions

Tushar: conceptualization, methodology, validation, data curation, formal analysis, investigation, writing—original draft, writing—review and editing; **Charles Granger**: data curation, methodology, formal analysis, writing—original draft, writing—review and editing; **Ali Altamimi**: data curation, formal analysis, methodology, writing—original draft, writing—review and editing; **Yoel Cortes-Pena**: methodology, writing—review and editing; **Kevin Nelson**: investigation, data curation, writing—review and editing; **Xiaobo Dong**: investigation, writing—review and editing; **Karl Britt**: investigation, writing—review and editing; **Andrea Thurnheer**: investigation, writing—review and editing; **Louis Barrows**: investigation, writing—review and editing; **Styliani Avraamidou**: conceptualization, funding acquisition, writing—review and editing; **Reid C. Van Lehn**: conceptualization, funding acquisition, supervision, writing—review and editing; **George W. Huber**: conceptualization, formal analysis, funding acquisition, project administration, supervision, writing—review and editing.

Notes

The authors declare no competing financial interest.

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